Name_____ Date_____ Period_____

LESSON 7-2 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Rational Numbers on a Number Line

Maryland Standard 6.NS.C.6c · Learning target – I can plot a rational number and its opposite on a number line.

REMEMBER

- A **rational number** can be written as a fraction a/b of two integers ($b \neq 0$). Examples: 3/4, −2.5, 7, −1 1/2.
- Between two integers, split the gap into **equal parts** (halves, fourths, tenths).
- Negative numbers sit **left of 0**. −2.5 is halfway between −3 and −2.

WORKED EXAMPLE

Plot −1.75 and its opposite on a number line.

Step 1 −1.75 is between −2 and −1. Split each unit into fourths; count 3 fourths left of −1.

Step 2 The opposite, 1.75, is 3 fourths right of 1.

Answer **points at −1.75 and 1.75**

WORD BANK

rational number – a number you can write as a fraction

integer – …, −2, −1, 0, 1, 2, …

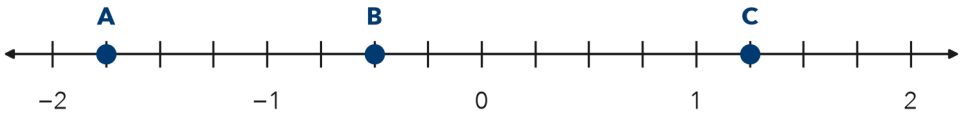
negative – less than 0; to the left of 0

between – in the middle of two numbers

PRACTICE

show your work · label units

1. Write the number for each point as a decimal or a fraction.



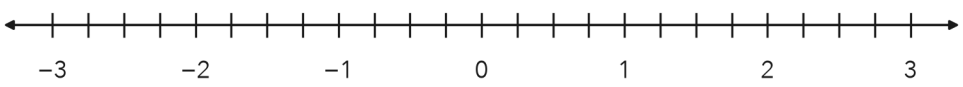
a. A = _____ **b.** B = _____ **c.** C = _____

2. Plot and label each number on the number line: −2.5, 1/2, −0.75, 1 1/4

Step 1 Find the two integers on each side of the number.

Step 2 Split the gap into equal parts.

Step 3 Count the parts and mark the point.



3. Name the two integers each number is between.

a. −4.6 is between _____

b. 7/2 is between _____

c. −0.3 is between _____

d. −9/4 is between _____

4. Write a rational number for each situation.

a. a loss of $3.50 ____

b. a gain of 2 1/2 ft ____

c. 4.5 degrees below 0 ____

5. **Pond.** A frog is 1.5 feet **above** the pond. A fish is 2.25 feet **below** the top of the pond. The top of the pond is 0.

Frame *The ___ is closer to 0 because ___.*

a. Write a number for the frog and for the fish. _____

b. Which one is closer to the surface? _____

6. **Find the Error.** Tyrell plotted −1.5 between −1 and 0. Explain his mistake.

Frame *Tyrell's mistake is ___. −1.5 is between ___ and ___.*

_____

_____

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___

AT HOME Find a ruler or measuring cup at home. Show someone where 1/2, 1/4, and 3/4 are.

Name_____ Date_____ Period_____

LESSON 7-2 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Opposites of Rational Numbers

Maryland Standard 6.NS.C.6a · Learning target – I can find the opposite of a rational number and explain what taking the opposite twice does.

REMEMBER

- **Opposites** are the same distance from 0: 2.5 and −2.5.
- The opposite of a negative number is **positive**.
- Taking the opposite **twice** gives back the number you started with.

WORKED EXAMPLE

Start at 3/5. Find its opposite. Then find the opposite of that.

Step 1 Opposite of 3/5 is −3/5.

Step 2 Opposite of −3/5 is 3/5.

Answer **−3/5, then 3/5**

WORD BANK

opposite – same distance from 0, other side

−(−3) – the opposite of −3, which is 3

distance from 0 – how far a number is from 0

PRACTICE

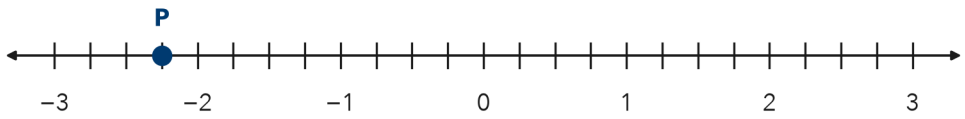
show your work · label units

1. Write the opposite of each number, or find the value.

a. 4.8 _____ **b.** −2/3 _____ **c.** −0.05 _____

d. 0 _____ **e.** −(−1.6) = _____ **f.** −(−5/8) = _____

2. Point P is shown. Plot its opposite and label it Q.



Q = _____

3. **Game Score.** Lena has −12.5 points. A bonus card changes her score to the opposite number.

What is her new score? How far from 0 is it? _____

4. Stretch. Point M is 3.75 units from 0, to the left of 0. Point N is the opposite of M.

Step 1 M is to the left of 0, so M is negative.

Step 2 N is the opposite of M, so N is positive.

a. M = _____

b. N = _____

5. Find the Error. Kofi says −(−2/5) = −2/5. Explain his mistake. Give the correct value.

Frame *Kofi is wrong because the opposite of a negative number is ___.*

_____

_____

TEST PRACTICE

choose the best answer

6. What is the opposite of −7/8?

A −7/8 **B** 7/8 **C** 8/7 **D** −8/7

7. Point R is at 3.4. Where is the opposite of the opposite of R?

A −3.4 **B** 0 **C** 3.4 **D** 4.3

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Explain to someone at home why −(−1.5) is a positive number.